

PIRP / PIRU

Portable Intelligence Runtime Pieces and Units

From Runtime Intelligence to Relay Localization and Collective Computational Growth

"Can intelligence itself become a portable, relayable, structurally localizable, composable, and computationally growable runtime object?"

CORE IDEA

PIRP preserves discovery-oriented runtime intelligence before complete computational classification. PIRU is its more formalized, bounded, typed, validated, and reusable engineering form. Together they provide a vocabulary for intelligence that can move across humans, AIs, tools, runtimes, organizations, domains, and generations.

PIRP - Portable Intelligence Runtime Piece A bounded-enough piece of intelligence that can be identified, carried, published, received, localized, continued, executed, observed, validated, composed, relayed, and grown. Principle: Intelligence precedes its computational classification.	Open PIRP - Unresolved Intelligence A well-structured problem can itself be Portable Intelligence. An Open PIRP can preserve Known, Unknown, the Known-Unknown Boundary, structural gaps, evidence, counter-evidence, failed attempts, required capabilities, tools, validation conditions, and open interfaces. Principle: A validated failure can be positive intelligence.
PIRU - Portable Intelligence Runtime Unit A more formalized Portable Intelligence structure with clearer identity, boundary, type, interface, runtime contract, validation, policy, and version.	Publishing as Intelligence Transfer Publishing can inject a Portable Intelligence structure into a larger intelligence runtime. A paper, DOI repository, issue, benchmark, experiment, or research question may become a continuation point for another human or AI.

CANONICAL GROWTH LOOP

Runtime Intelligence	-	PIRP	-	PIRU	-	Publish / Relay	-	Structural Delta	-	Collective Growth
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RELAY LOCALIZATION: A SECOND FORM OF LOCALIZATION

Structural Localization Where is relevant existing intelligence?	Relay Localization Who or what should continue this intelligence?
Knowledge Distance Distance to relevant existing intelligence.	Capability + Relay Distance Fit for continuation and cost of productive transfer.

Key principle: Localization is not merely finding where an answer already exists. It is finding where a piece of intelligence can most productively continue to exist and grow. **Social Proximity is not Intelligence Proximity.**

COLLECTIVE INTELLIGENCE CALLINGGRAPH

Knowledge CallingGraph records what intelligence depends on, extends, validates, contradicts, or composes with other intelligence. **Relay CallingGraph** records how intelligence actually moved, who or what received it, under what context, and what Structural Delta followed. Together they preserve not only knowledge, but the history of how knowledge grew.

Every relay can teach the system how to relay better next time. Collective Learning therefore becomes both Knowledge Growth and Relay Intelligence Growth.

EIGHT CORE RESEARCH ARTICLES

- 001 - From Per-Node Intelligence to Portable Runtime Intelligence
- 002 - Pre-Categorical Intelligence and the PIRP/PIRU Duality
- 003 - PIRP Anatomy, Lifecycle, and Computational Growth
- 004 - Publishing as Portable Intelligence Transfer
- 005 - Open PIRPs and Unresolved Intelligence
- 006 - Relay Context and the Collective Intelligence CallingGraph
- 007 - Relay Localization and Computational Growth Dispatch
- 008 - The PIRP/PIRU Ecology and an Evolutionary Roadmap

v1.0.0 DESIGN BOUNDARY

This release establishes the conceptual and structural framework. It intentionally does not prescribe a mandatory Java runtime, agent framework, central registry, serialization format, or universal PIRP schema. Future engineering directions include PIRP Metadata, Open-PIRP Exchange, Relay Graph, Structural Matcher, Relay Localization Engine, PIRP Dispatch Plane, AI PIRP Extraction, and AI-generated Open PIRPs.